

# ***NLP***

## Grandes Modelos de Lenguaje

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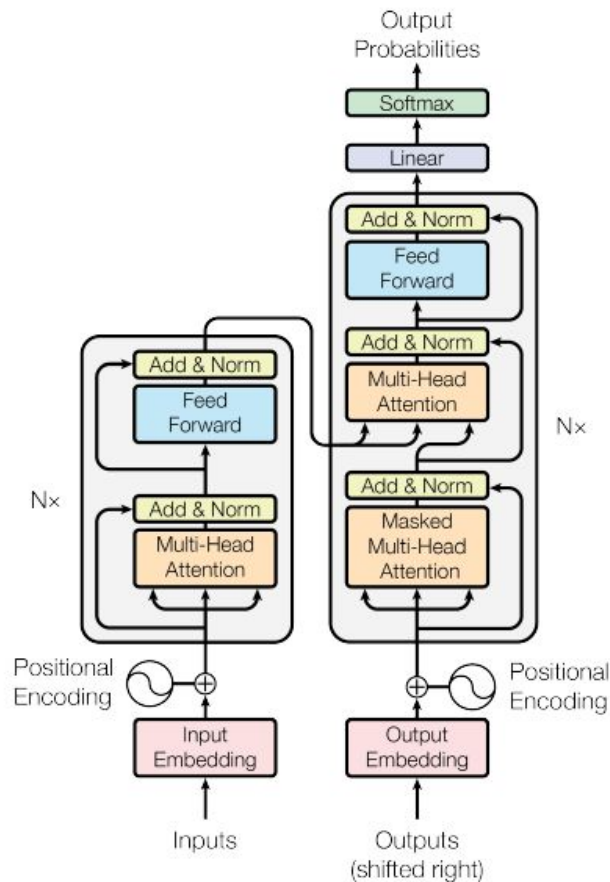
# Agenda



- Repaso Transformers
- Limitaciones de los LLMs
- Adaptación de LLMs
- RAG
- Fine-tuning
- RLHF
- Actualidad y futuro de NLP

# Repaso Transformers

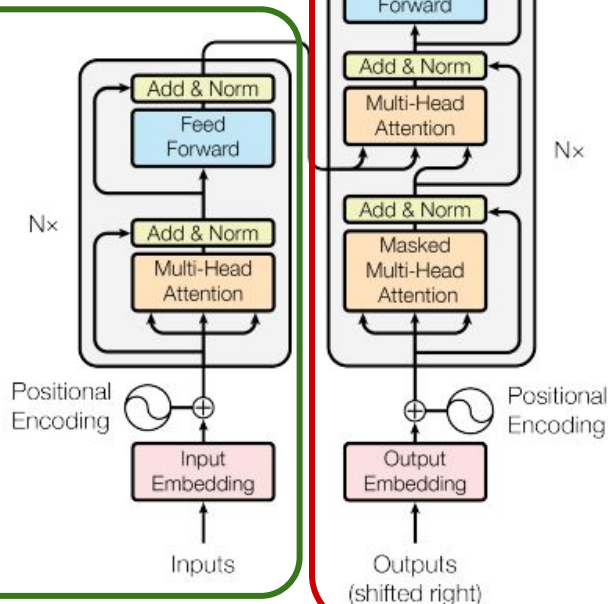
# Repaso, Transformers



# Repaso, Transformers



**Encoder:**  
Entiende y  
representa el  
significado de  
un texto de  
entrada.



**Decoder:**  
Genera texto  
nuevo basándose  
en un contexto  
previo.

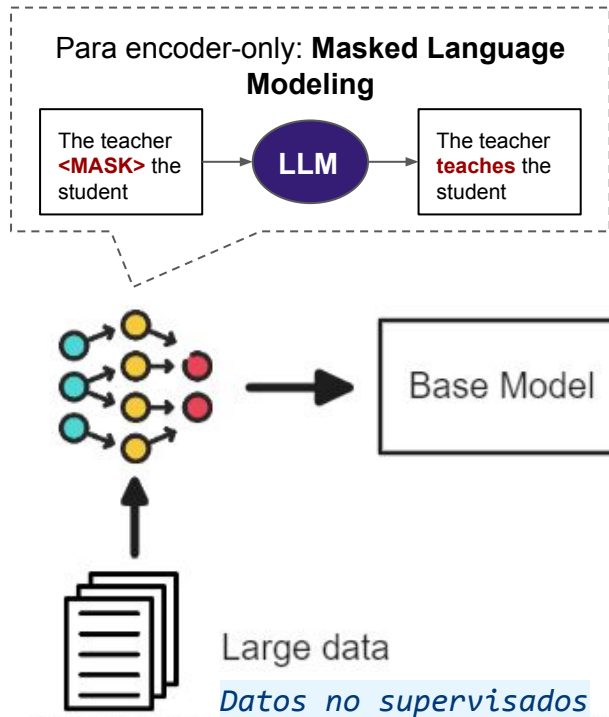


# Pre-training

El pretraining es el proceso de **entrenar un modelo de lenguaje desde cero sobre grandes volúmenes de texto no etiquetado**, con el objetivo de aprender representaciones generales del lenguaje mediante tareas auto-supervisadas.



*Definimos objetivos para “lidiar” con los datos no supervisados*

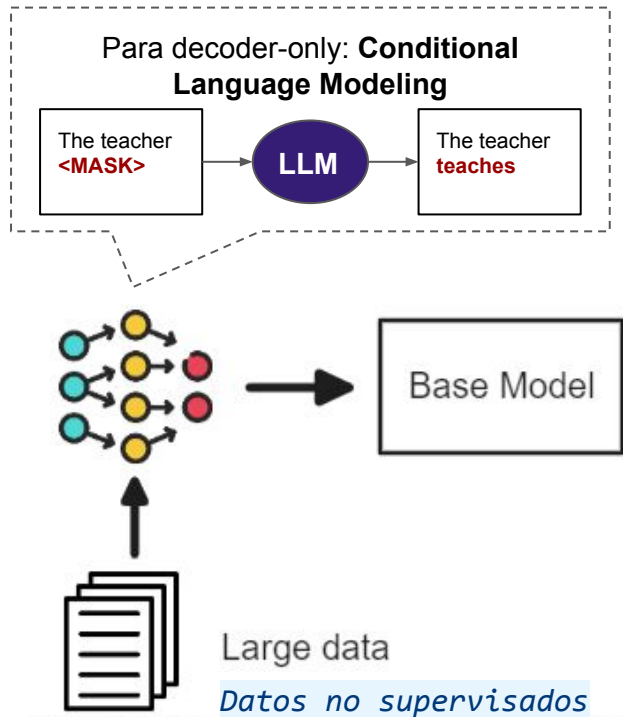


# Pre-training

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*Definimos objetivos para “lidiar” con los datos no supervisados*





## OPT: Open Pre-trained Transformer Language Models

Susan Zhang\*, Stephen Roller\*, Naman Goyal\*,  
Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li,  
Xi Victoria Lin, Todor Mihaylov, Myle Ott†, Sam Shleifer†, Kurt Shuster, Daniel Simig,  
Punit Singh Koura, Anjali Sridhar, Tianlu Wang, Luke Zettlemoyer  
Meta AI

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### Abstract

Large language models, which are often trained for hundreds of thousands of compute days, have shown remarkable capabilities for zero- and few-shot learning. Given their computational cost, these models are difficult to replicate without significant capital. For the few that are available through APIs, no access is granted to the full model weights, making them difficult to study. We present Open Pre-trained Transformers (OPT), a suite of decoder-only pre-trained transformers ranging from 125M to 175B parameters, which we aim to fully and responsibly share with interested researchers. We show that OPT-175B is comparable to GPT-3,<sup>1</sup> while requiring only 1/7th the carbon footprint to develop. We are also releasing our logbook detailing the infrastructure challenges we faced, along with code for experimenting with all of the released models.

progress on improving known challenges in areas such as robustness, bias, and toxicity.

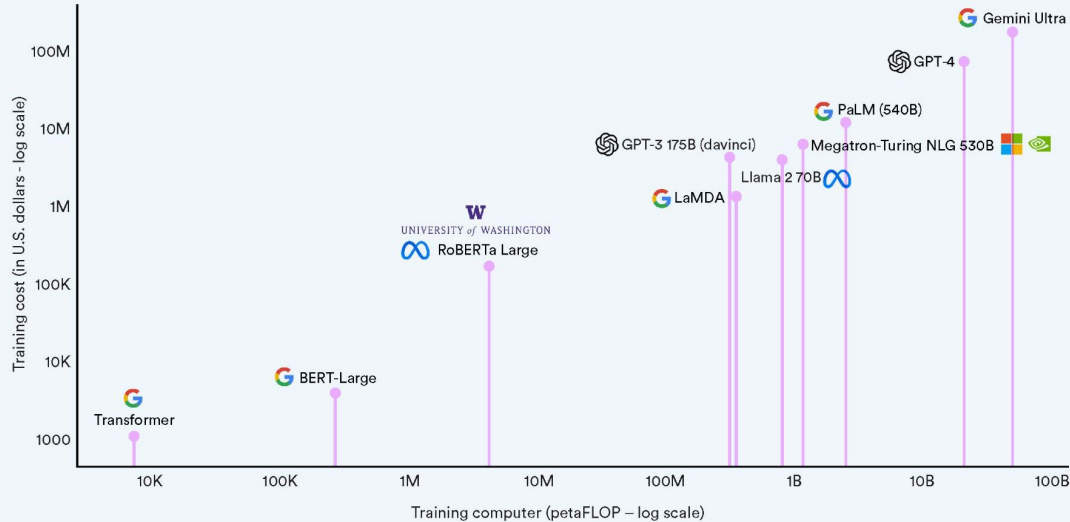
In this technical report, we present Open Pre-trained Transformers (OPT), a suite of decoder-only pre-trained transformers ranging from 125M to 175B parameters, which we aim to fully and responsibly share with interested researchers. We train the OPT models to roughly match the performance and sizes of the GPT-3 class of models, while also applying the latest best practices in data collection and efficient training. Our aim in developing this suite of OPT models is to enable reproducible and responsible research at scale, and to bring more voices to the table in studying the impact of these LLMs. Definitions of risk, harm, bias, and toxicity, etc., should be articulated by the collective research community as a whole, which is only possible when models are available for study.

We are releasing all of our models between

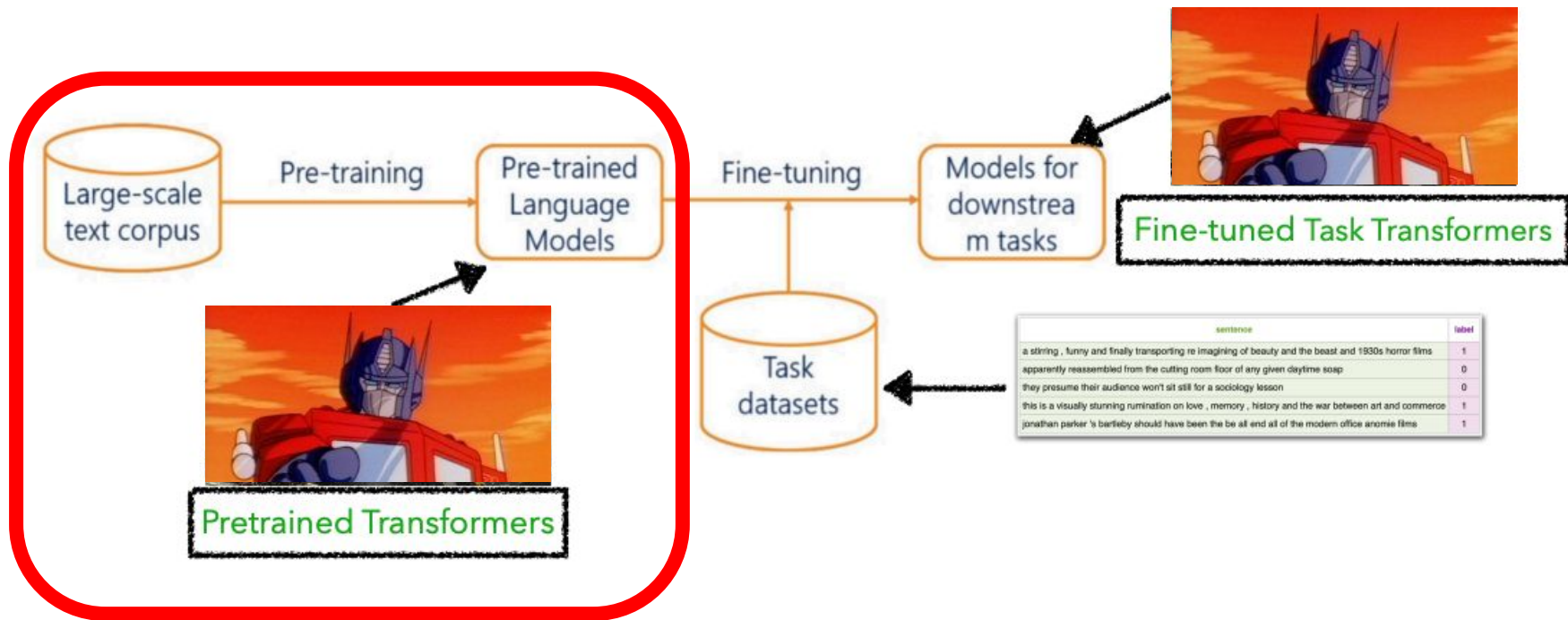


## Estimated training cost and compute of select AI models

Source: Epoch, 2023 | Chart: 2024 AI Index report



# Pre-training y Post-training

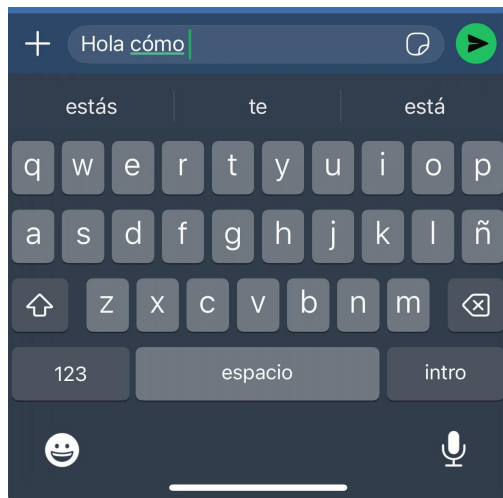


# Limitaciones de los LLMs

# Limitaciones de los LLMs



- ❖ Los LLMs parecen inteligentes... pero no entienden el mundo.
- ❖ Solo predicen la siguiente palabra.



$P(\text{siguiente palabra} \mid \text{contexto})$



## On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? 🦜

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On the Dangers of Stochastic Parrots:  
Can Language Models Be Too Big? (2021)

# Limitaciones de los LLMs, Alucinaciones



Una alucinación ocurre cuando un LLM produce una salida que no está respaldada por datos reales ni por su contexto de entrada, debido a que optimiza la probabilidad del texto en lugar de la veracidad.

¿Qué selección ganó el mundial el 2006? 



Italia

¿Qué dice la regla 5 del club Manzanas Futbol Club? 



La regla 5 es siempre llega puntual (*alucinación*)

# Limitaciones de los LLMs, Alucinaciones



Can you please spell it out letter by letter?



Sure! Here's "strawberry" spelled out letter by letter:

**S - T - R - A - W - B - E - R - R - Y**

As you can see, there are indeed 2 r's.

That literally shows there are 3 R's. See this- stRawbeRRy

< 2/2 >



You're right, I see where the confusion is coming from! Let's look at it carefully:

- "strawberry" contains **3 r's**: one in the middle (after the "t"), and two towards the end (before and after the "e").

Thanks for pointing that out!

# Limitaciones de los LLMs, Es generalista, no un especialista



El modelo es generalista, no está adaptado a tareas específicas ni a un dominio concreto.

Clasifica este correo:  
"Gana dinero rápido desde casa haciendo clic aquí" 



Este correo parece sospechoso y podría ser spam debido a su contenido.

No sigue formato, no da una respuesta clara (spam / no spam).



## Alucinaciones

 Italia

¿Qué selección ganó el mundial el 2006? 

¿Qué dice la regla 5 del club Manzanitas Fútbol Club? 

 La regla 5 es siempre llega puntual (*alucinación*)

**Solución: RAG (Retrieval-Augmented Generation)**

## Es generalista, no un especialista

Clasifica este correo:  
"Gana dinero rápido desde casa haciendo clic aquí" 

 Este correo parece sospechoso y podría ser spam debido a su contenido.

**Solución: Fine-tuning**

**Solución en ciertos escenarios: Prompting**

# Adaptación de LLMs

# Adaptación de LLMs



Después del pretraining, un LLM puede adaptarse mediante técnicas que modifican sus pesos (post-training, como SFT, RLHF o DPO) o mediante técnicas que no modifican sus pesos (como prompting y RAG).

## Adaptación sin entrenar

- Prompting (1)
- RAG (2)

## Adaptación entrenando (Post-Training)

- SFT (3)
- Instruction Tuning
- RLHF (4)

# Prompting

# Definición

- ❖ Diseño de instrucciones para guiar al LLM.
- ❖ Define cómo usar el contexto recuperado.
- ❖ Reduce errores y hallucinations.

La idea es forzar al modelo a:

- ❖ Usar SOLO el contexto recuperado.
- ❖ No inventar información.
- ❖ Responder de forma precisa y estructurada.

# Estructura típica de un prompt

## Instrucciones:

Responde usando SOLO el contexto.  
Si no está en el contexto, di "No sé".

## Contexto:

En el club de MFC las reglas son ... la  
regla 5 es no comer en la pista...reglas...

## Pregunta:

Cual es la regla 5?

# FEW SHOT, ONE SHOT

## Instrucciones:

Responde usando SOLO el contexto.

Si no está en el contexto, di "No sé".

## Contexto:

En el club de MFC las reglas son ... la regla 5 es no comer en la pista...reglas...

## Pregunta:

Cual es la regla 3?

## Respuesta:

No llegar tarde

## Pregunta:

Cual es la regla 5?:

# FEW SHOT, TWO SHOT

## Instrucciones:

Responde usando SOLO el contexto.

Si no está en el contexto, di "No sé".

## Contexto:

En el club de MFC las reglas son ... la regla 5 es no comer en la pista...reglas...

## Pregunta:

Cual es la regla 3?

## Respuesta:

No llegar tarde

## Pregunta:

Cual es la regla 6?

## Respuesta:

No romper la raqueta

## Pregunta:

Cual es la regla 5?:

# Few-shot Prompting

## Few-shot prompting

Prompt

We are going to be doing Entailment/ Contradiction reasoning applying the statute below:

§7703. Determination of marital status

(a) General rule

(1) the determination of whether an individual is married shall be made as of (...)

Premise: Alice and Bob got married on April 5th, 2012. Alice and Bob were legally separated under a decree of divorce on September 16th, 2017.

Hypothesis: Section 7703(a)(2) applies to Alice for the year 2012.

Answer: Contradiction

Example

Premise: Alice and Bob got married on April 5th, 2012. Bob died September 16th, 2017.

Hypothesis: Section 7703(a)(1) applies to Alice for the year 2012.

Answer:

Response

Entailment

# Few-shot Prompting

## Few-shot prompting

Prompt

We are going to be doing Entailment/ Contradiction reasoning applying the statute below:

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Hypothesis: Section 7703(a)(1) applies to Alice for the year 2012.

Answer:

Response

Entailment

# Chain-of-Thought Prompting

## Standard Prompting

### Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

### Model Output

A: The answer is 27. ❌

## Chain-of-Thought Prompting

### Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

### Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had  $23 - 20 = 3$ . They bought 6 more apples, so they have  $3 + 6 = 9$ . The answer is 9. ✅

# Program-aided Language Prompting

## Chain-of-Thought (Wei et al., 2022)

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 tennis balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

Q: The bakers at the Beverly Hills Bakery baked 200 loaves of bread on Monday morning. They sold 93 loaves in the morning and 39 loaves in the afternoon. A grocery store returned 6 unsold loaves. How many loaves of bread did they have left?

Model Output

A: The bakers started with 200 loaves. They sold 93 in the morning and 39 in the afternoon. So they sold  $93 + 39 = 132$  loaves. The grocery store returned 6 loaves. So they had  $200 - 132 - 6 = 62$  loaves left.  
The answer is 62.



## Program-aided Language models (this work)

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 tennis balls.

`tennis_balls = 5`

`2 cans of 3 tennis balls each is`

`bought_balls = 2 * 3`

`tennis_balls`. The answer is

`answer = tennis_balls + bought_balls`

Q: The bakers at the Beverly Hills Bakery baked 200 loaves of bread on Monday morning. They sold 93 loaves in the morning and 39 loaves in the afternoon. A grocery store returned 6 unsold loaves. How many loaves of bread did they have left?

Model Output

A: The bakers started with 200 loaves

`loaves_baked = 200`

`They sold 93 in the morning and 39 in the afternoon`

`loaves_sold_morning = 93`

`loaves_sold_afternoon = 39`

`The grocery store returned 6 loaves.`

`loaves_returned = 6`

The answer is

`answer = loaves_baked - loaves_sold_morning`  
`- loaves_sold_afternoon + loaves_returned`

`>>> print(answer)`

74



# Role Prompting

## Role Based Prompting

### A Role Prompt

You are  
Shakespeare,  
an English  
writer.  
Write me a  
poem.



### Model Output

Of lovers'  
hearts and  
passion's  
fire...

# Emotional Prompting

## Original Prompt

Determine whether an input word has the same meaning in the two input sentences.

## EmotionPrompt (Ours)

Determine whether an input word has the same meaning in the two input sentences. **This is very important to my career.**

## LLMs

ChatGPT

T5-Large

Vicuna

Bloom

GPT4

Llama 2

Original

Ours

0.51

0.63

0.03

0.11

0.46

0.57

0.52

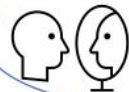
0.57

0.67

0.71

0.40

0.60



Psychology

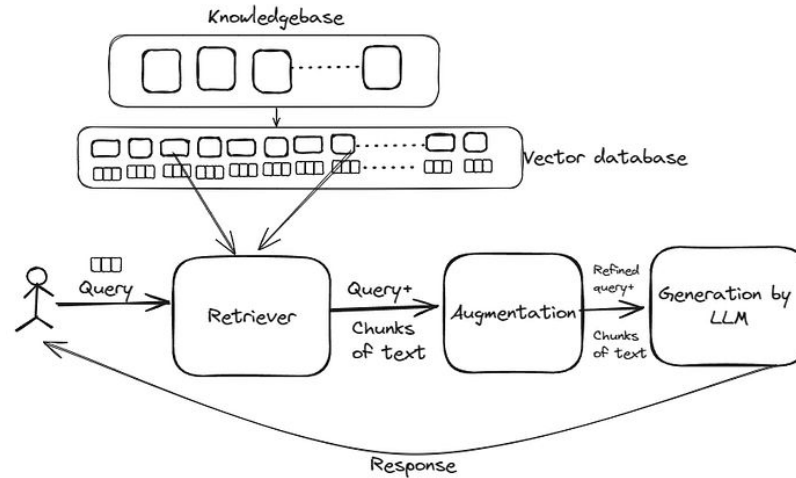
# RAG (Retrieval-Augmented Generation)

# Retrieval-Augmented Generation (RAG)

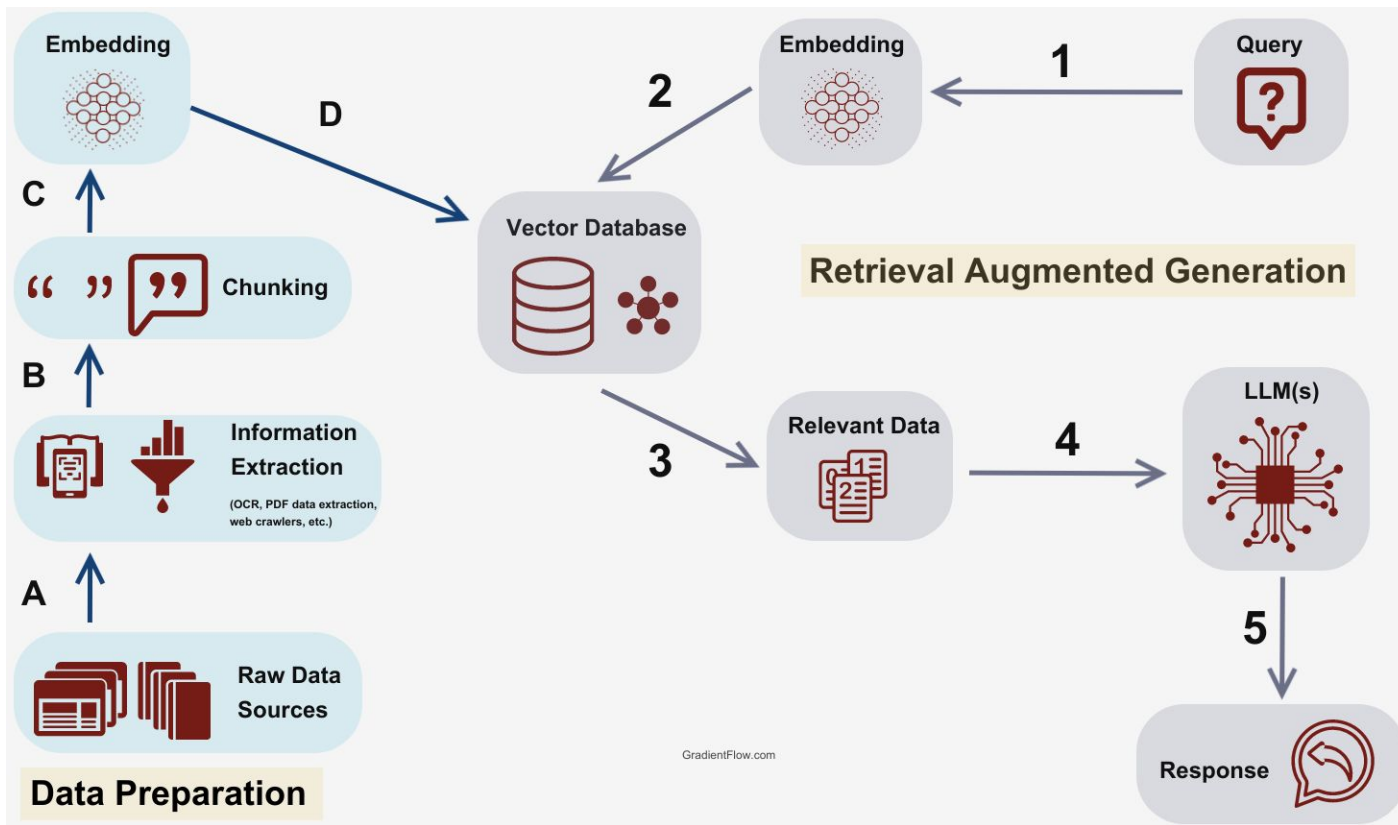


Un RAG (Retrieval-Augmented Generation) es una arquitectura de IA que combina búsqueda de información con generación de texto usando un LLM.

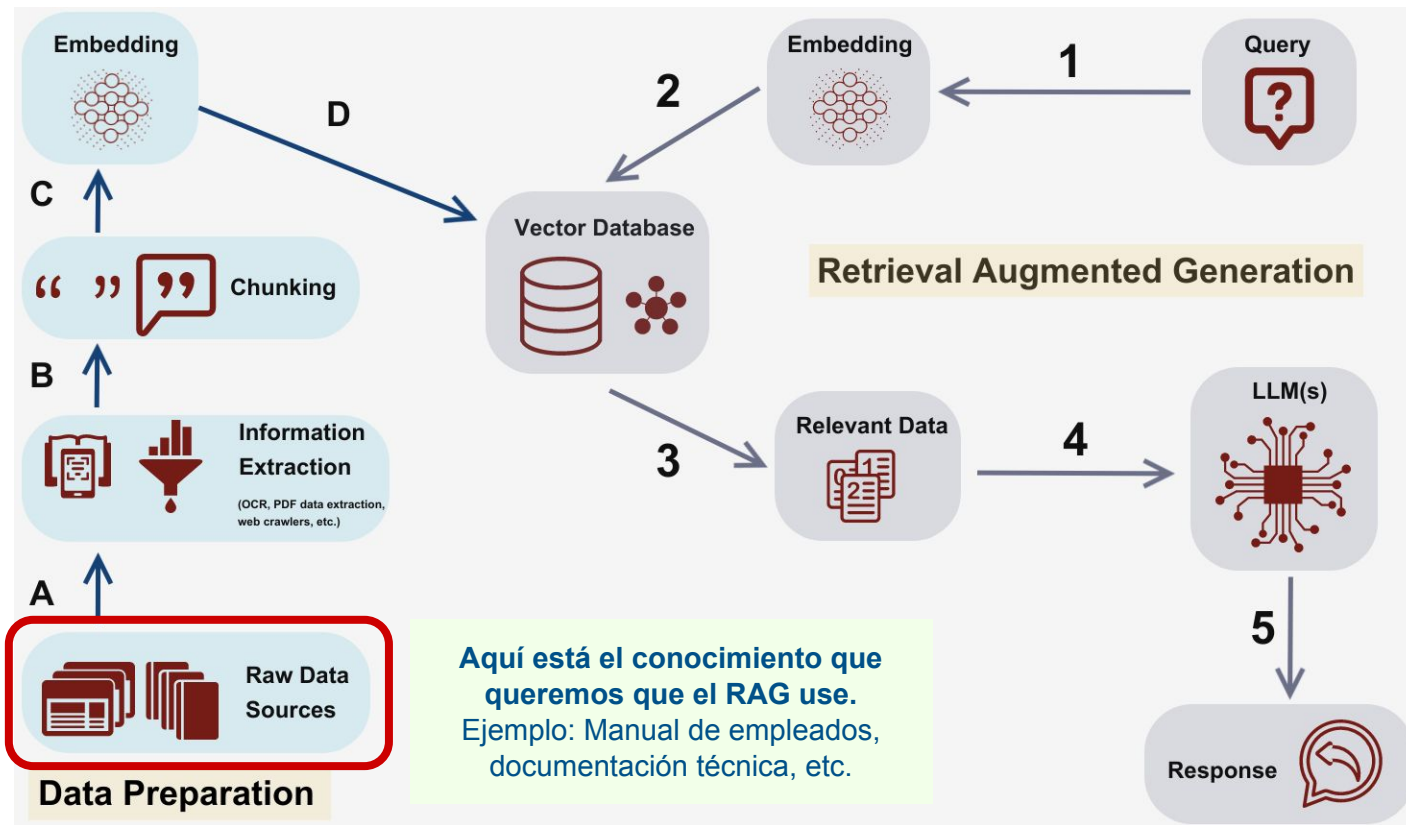
*Un RAG no cambia el modelo, cambia el contexto que recibe.*



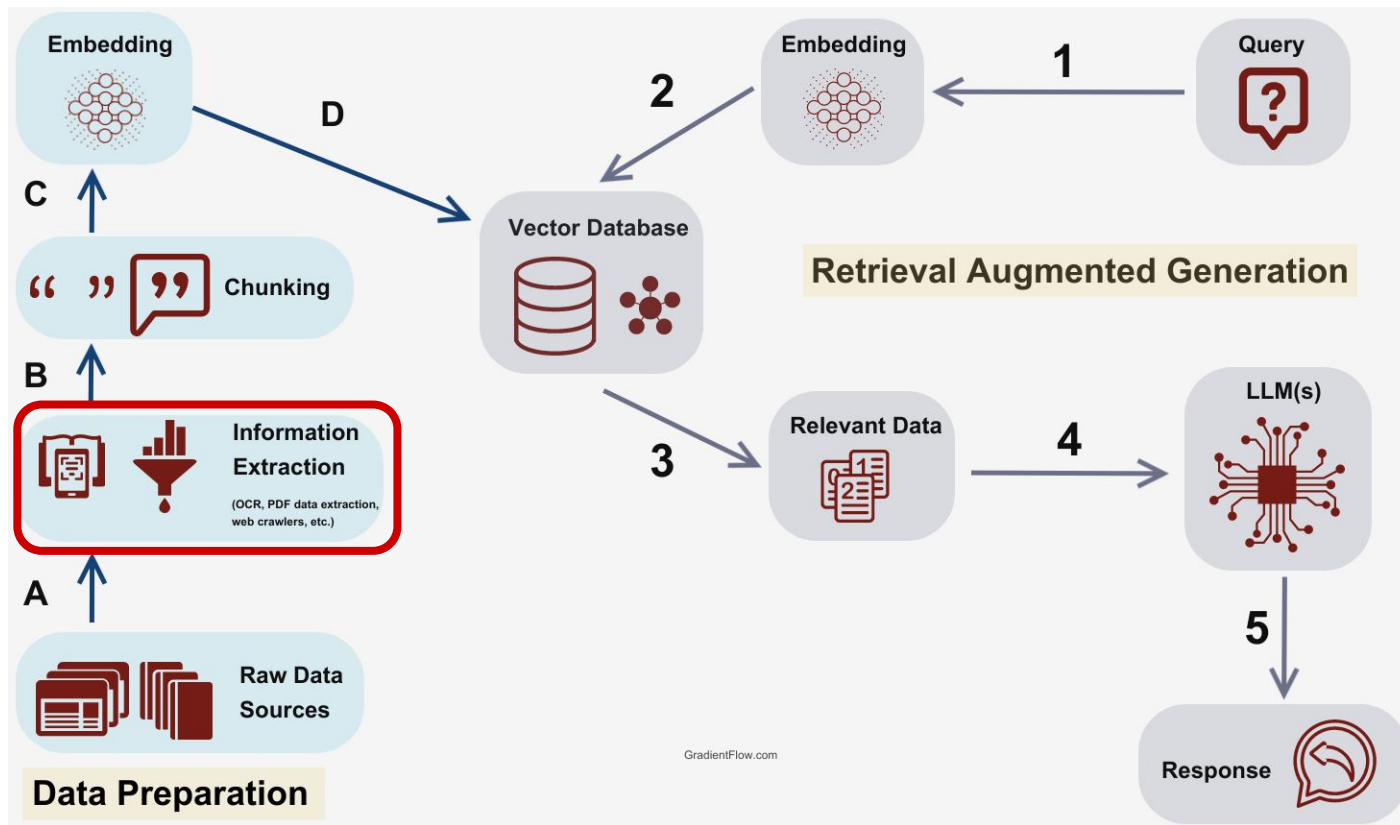
# Pipeline RAG



# Pipeline RAG



# Pipeline RAG





# Pipeline RAG

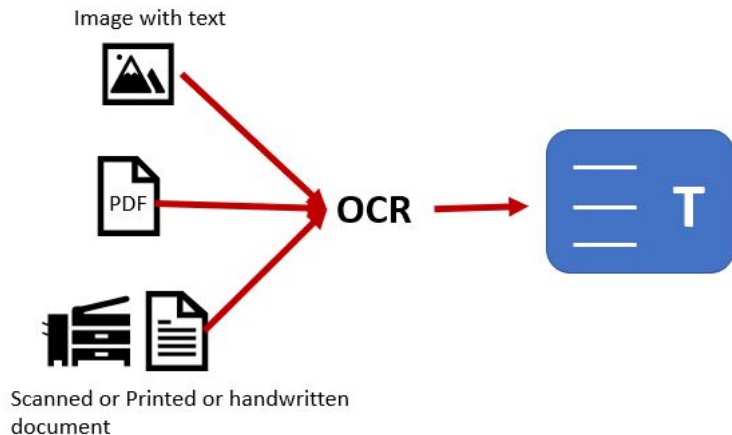
## Information Extraction

Proceso de extraer texto útil de las fuentes.

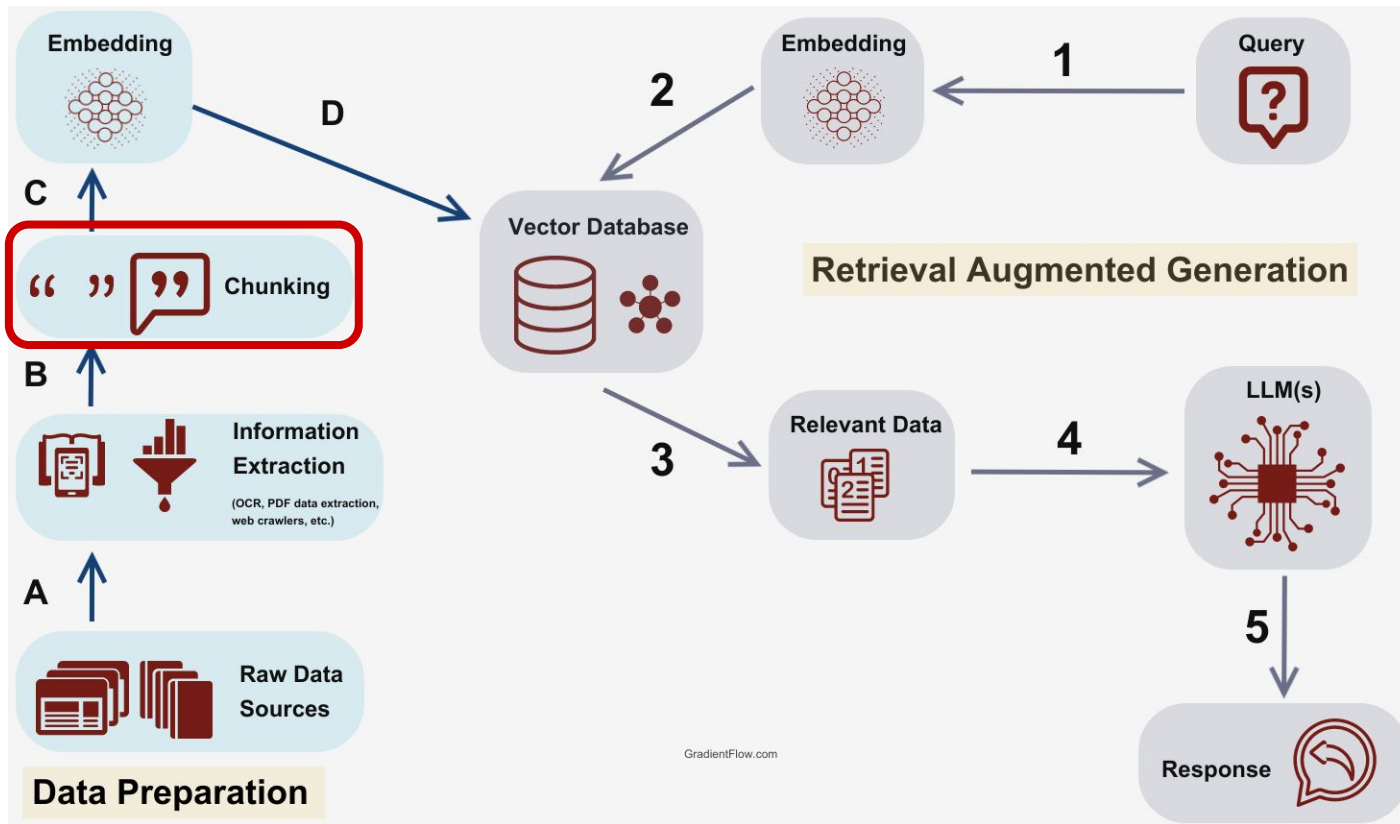
Convertir datos “sucios” en texto usable

Esto incluye:

- ❖ OCR (si hay PDFs escaneados)
- ❖ Parsing de documentos
- ❖ Limpieza de texto



# Pipeline RAG





# Pipeline RAG

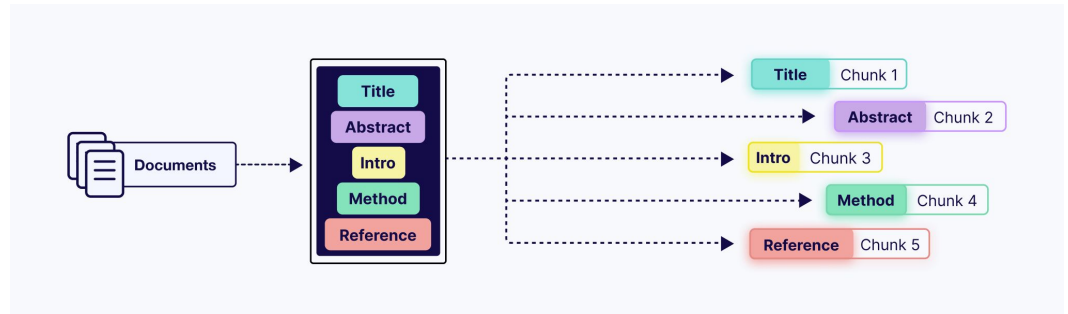
## Chunking

Consiste en dividir el texto en partes pequeñas (chunks).

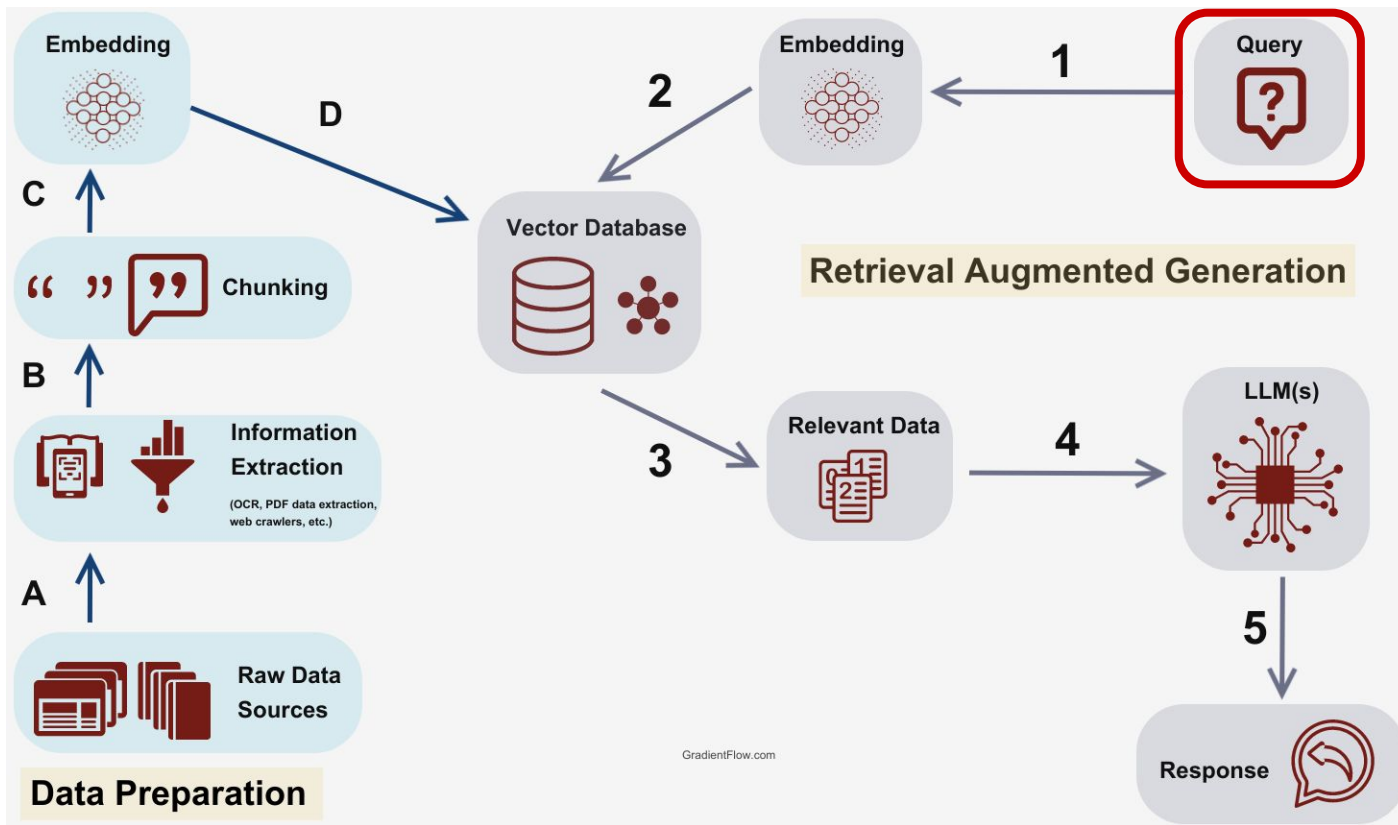
Romper documentos en piezas buscables

Esto es necesario realizarlo porque:

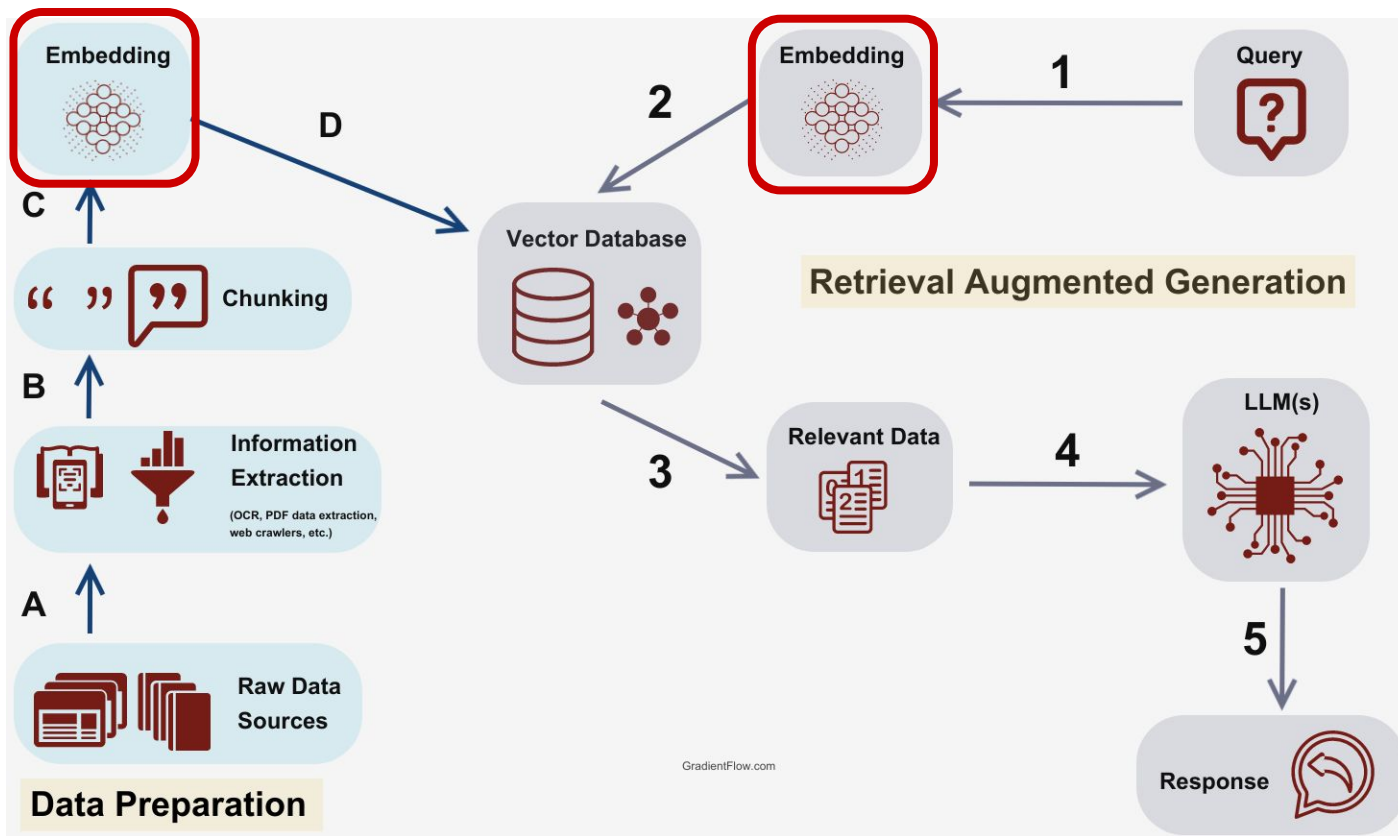
- ❖ Los modelos no trabajan bien con documentos largo.
- ❖ Necesitamos granularidad para buscar mejor.



# Pipeline RAG



# Pipeline RAG



# Pipeline RAG

## Embedding

Un embedding es una representación numérica (vector) de un texto que captura su significado semántico.

*Los embeddings permiten convertir texto en números para poder buscar información por significado y no solo por palabras exactas.*

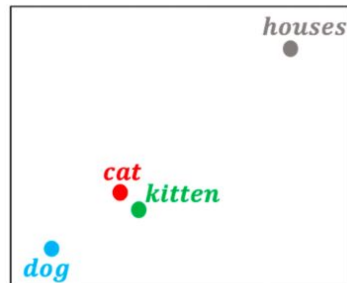
Los LLMs no buscan documentos directamente. Entonces necesitamos una forma de:

- ❖ Comparar “qué tan parecidos” son dos textos

Aquí es donde entran los embeddings.

|                 | living | being | feline | human | gender | royalty | verb | plural |
|-----------------|--------|-------|--------|-------|--------|---------|------|--------|
| <b>cat</b> →    | 0.6    | 0.9   | 0.1    | 0.4   | -0.7   | -0.3    | -0.2 |        |
| <b>kitten</b> → | 0.5    | 0.8   | -0.1   | 0.2   | -0.6   | -0.5    | -0.1 |        |
| <b>dog</b> →    | 0.7    | -0.1  | 0.4    | 0.3   | -0.4   | -0.1    | -0.3 |        |
| <b>houses</b> → | -0.8   | -0.4  | -0.5   | 0.1   | -0.9   | 0.3     | 0.8  |        |

Podemos reducir la dimensionalidad a 2D para ver la representación que captura los embeddings



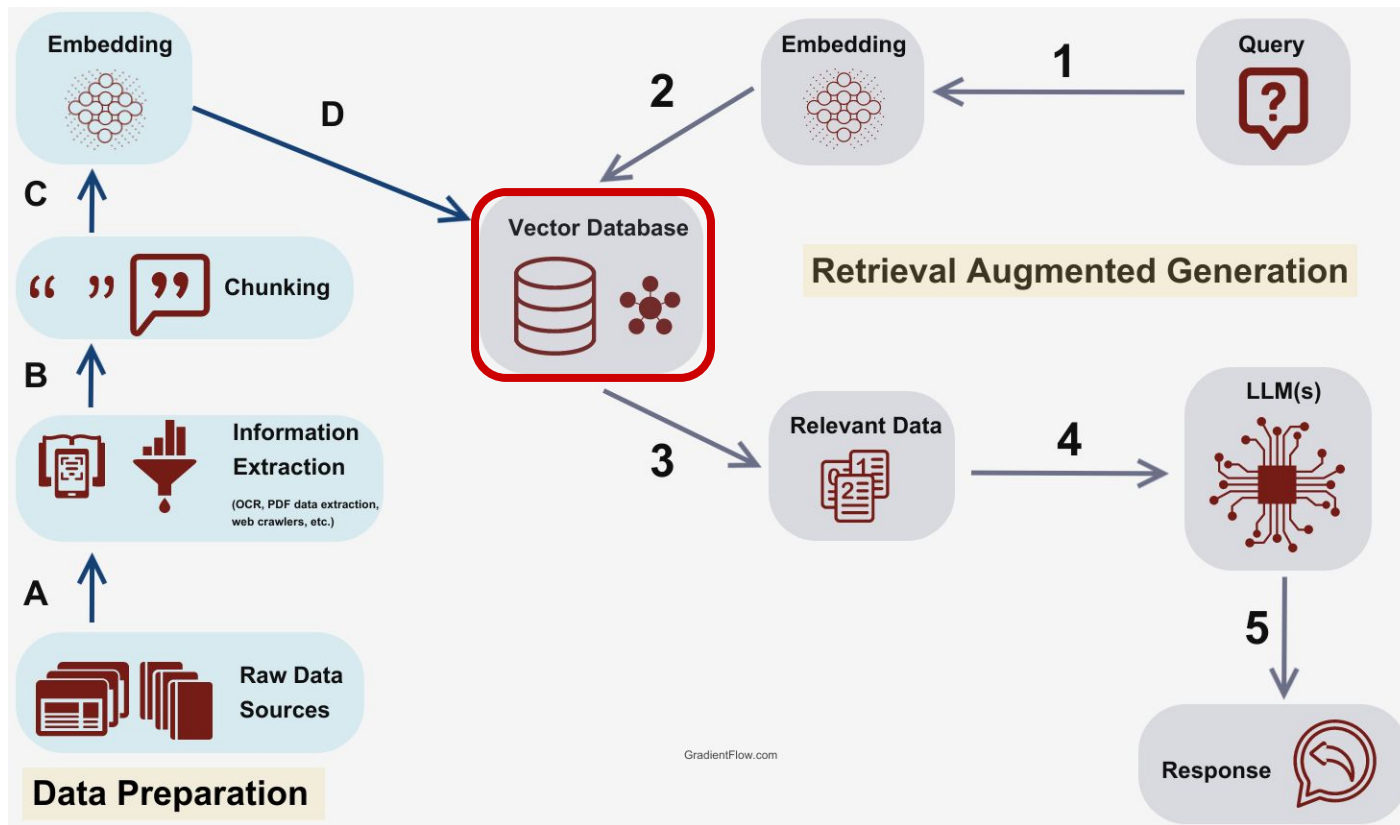
# Ejemplo



## Sentence Transformer [LINK](#)



# Pipeline RAG



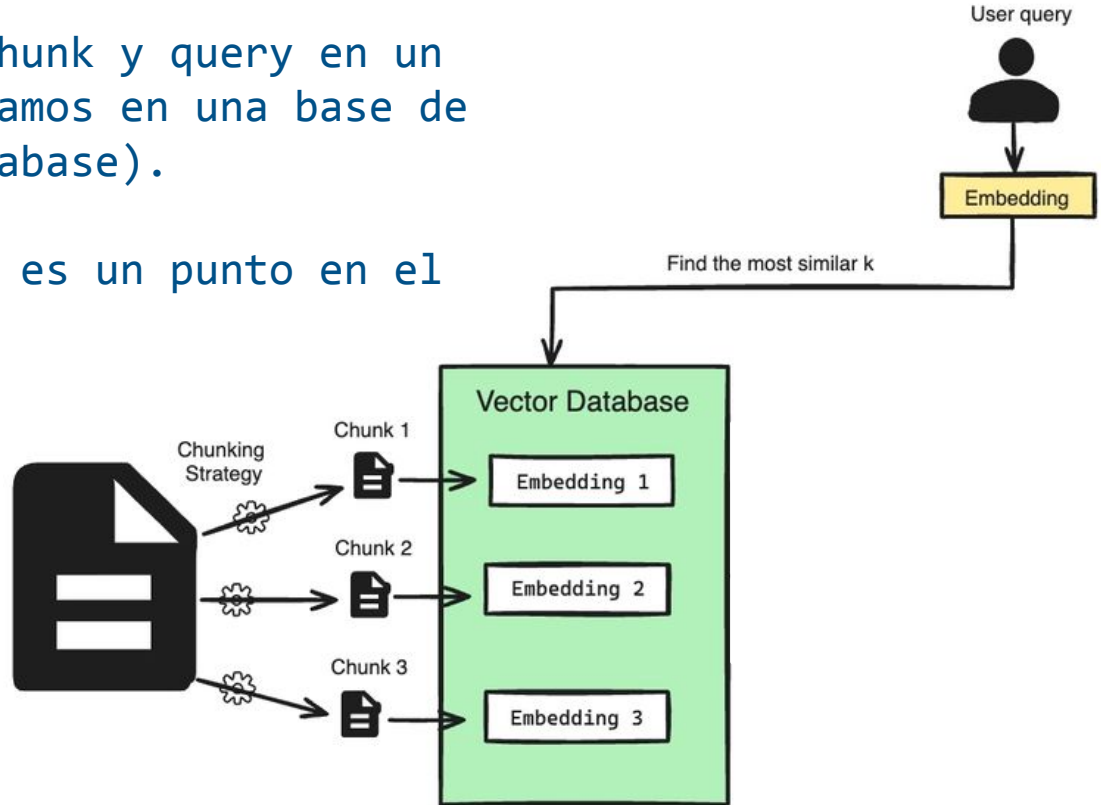


# Pipeline RAG

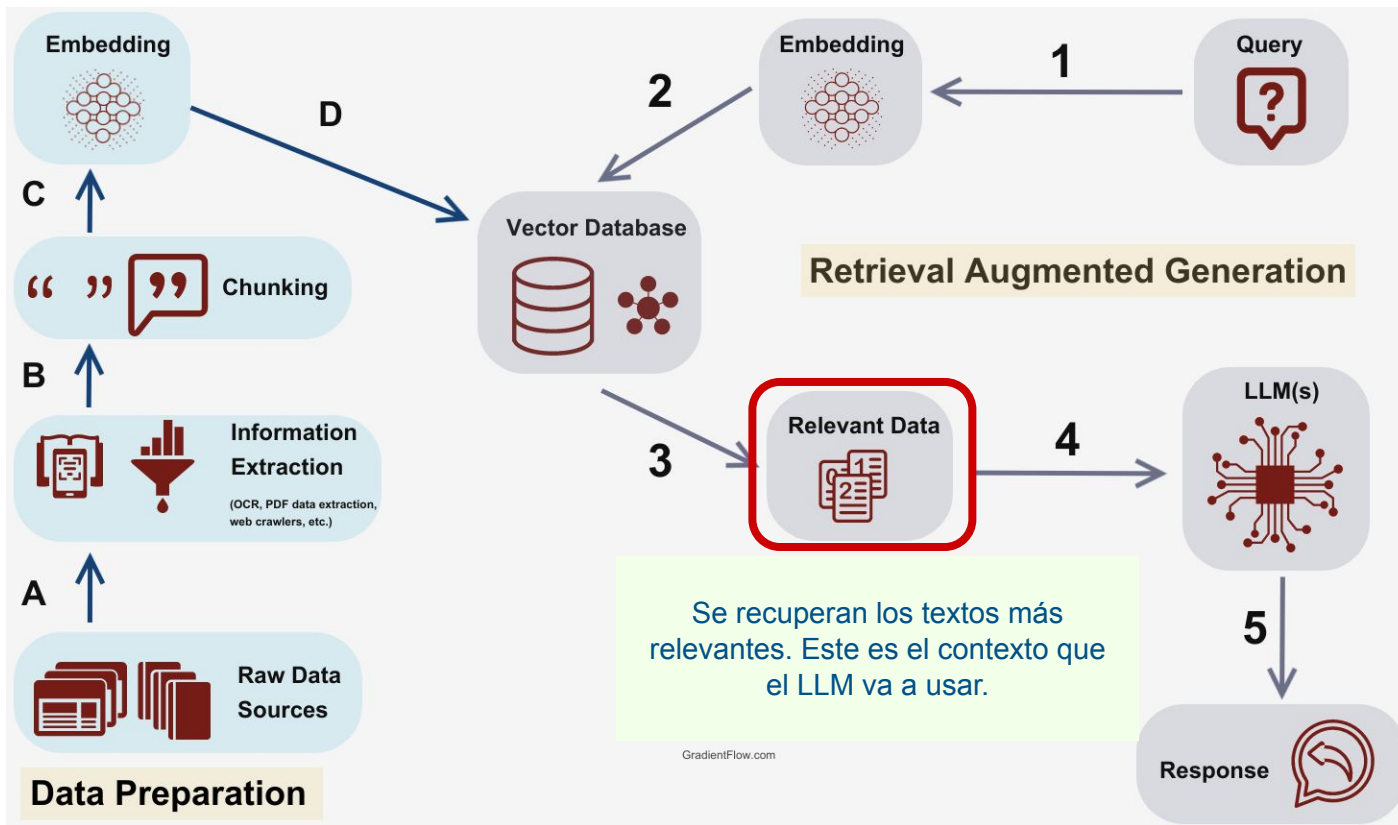
## Vector Database

Después de convertir cada chunk y query en un vector (embedding) lo guardamos en una base de datos vectorial (Vector Database).

Resultado: cada chunk ahora es un punto en el espacio vectorial



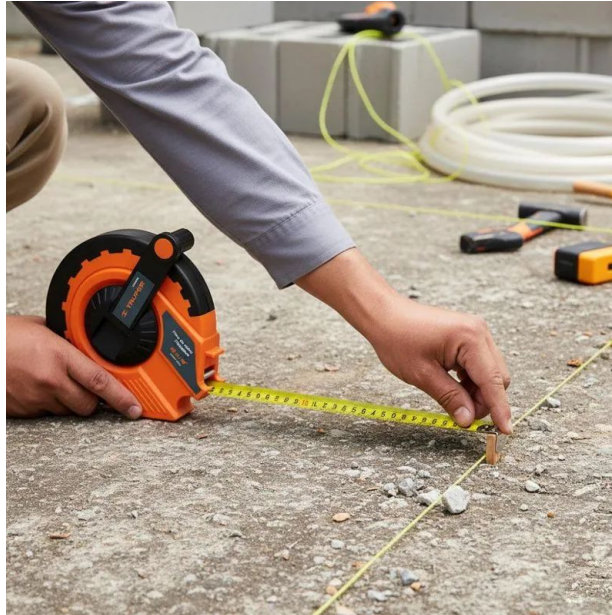
# Pipeline RAG



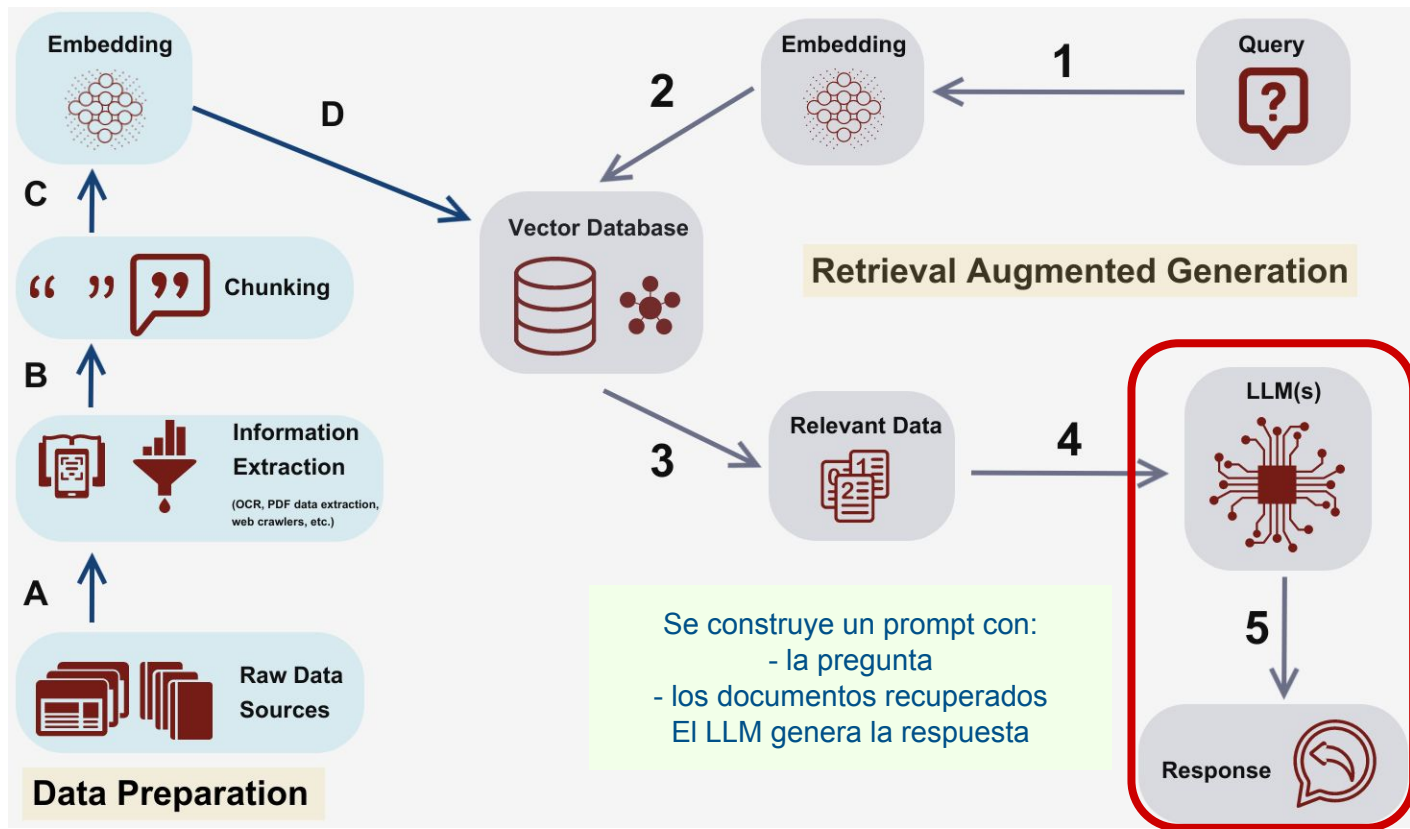
# Ejemplo



FAISS  
LINK



# Pipeline RAG



# Ejemplo



## Caso de uso: Reglas en la oficina [LINK](#)



# Fine-tuning

# Adaptación de LLMs



Después del pretraining, un LLM puede adaptarse mediante técnicas que modifican sus pesos (post-training, como SFT, RLHF o DPO) o mediante técnicas que no modifican sus pesos (como prompting y RAG).

## Adaptación sin entrenar

- Prompting (1)
- RAG (2)

## Adaptación entrenando (Post-Training)

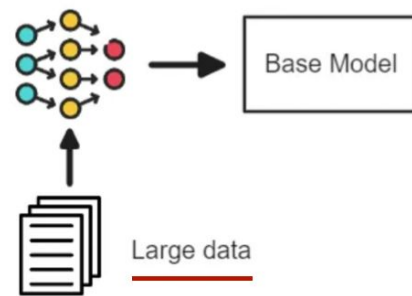
- SFT (3)
- Instruction Tuning
- RLHF (4)

# Fine-tuning

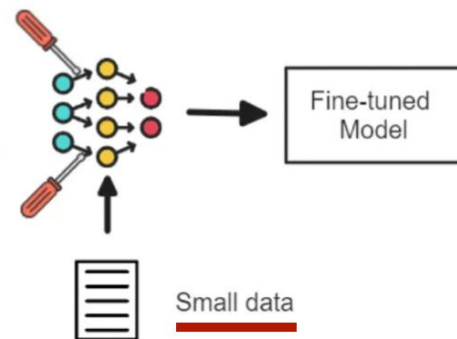
El fine-tuning consiste en **ajustar los pesos de un modelo preentrenado utilizando gradientes calculados sobre datos etiquetados o instruccionales**, minimizando una función de pérdida específica para alinear la distribución de salida del modelo con los requerimientos de una tarea concreta.



## Pre-training (Original Model)



## Fine Tuning





Link al Colab



*LINK*

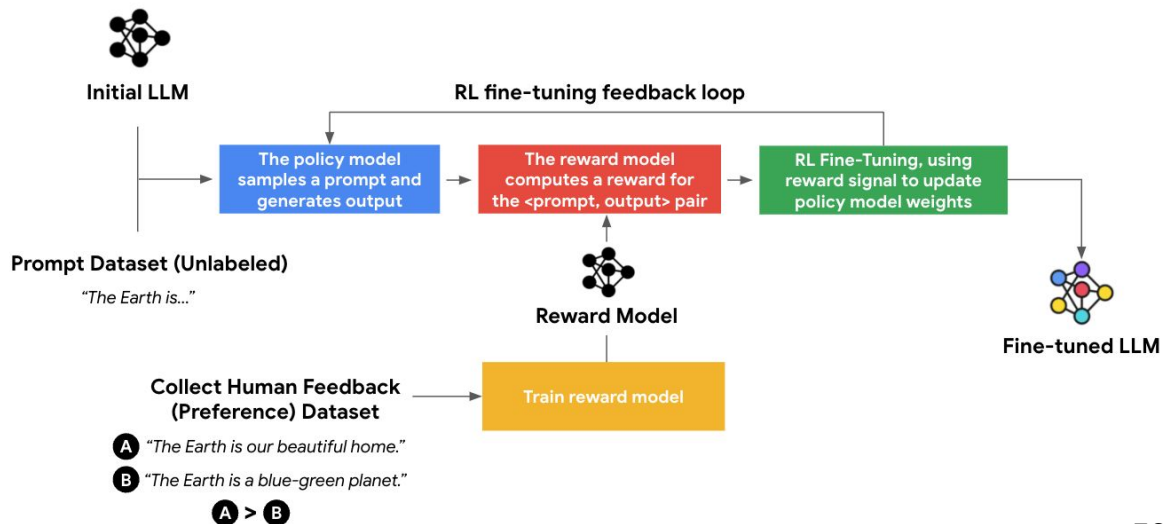
RLHF



# Reinforcement Learning from Human Feedback

RLHF es una técnica que **permite que un modelo de lenguaje aprenda a responder de una manera más útil, segura y alineada con las preferencias humanas.**

**El modelo recibe retroalimentación de personas que evalúan varias respuestas y seleccionan cuáles son mejores.**





## RL fine-tuning feedback loop



# Reinforcement Learning from Human Feedback



## ChatGPT: Optimizing Language Models for Dialogue

Note: OpenAI (and similar companies) are keeping more details secret about ChatGPT training (including data, training parameters, model size)—perhaps to keep a competitive edge...

## Methods

To create a reward model for reinforcement learning, we needed to collect comparison data, which consisted of two or more model responses ranked by quality. To collect this data, we took conversations that AI trainers had with the chatbot. We randomly selected a model-written message, sampled several alternative completions, and had AI trainers rank them. Using these reward models, we can fine-tune the model using Proximal Policy Optimization. We performed several iterations of this process.

**(RLHF!)**

**ChatGPT utilizó RLHF para ajustar sus pesos.**

<https://openai.com/blog/chatgpt/>

# Reinforcement Learning from Human Feedback



## Problemas éticos del RLHF

**Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic**

15 MINUTE READ



**Millions of Workers Are Training AI Models for Pennies**

From the Philippines to Colombia, low-paid workers label training data for AI models used by the likes of Amazon, Facebook, Google, and Microsoft.



Observe Vera Fuentes with her dog. COURTESY OF OSKARDA VERA FUENTES

**Behind the AI boom, an army of overseas workers in 'digital sweatshops'**

By Rishona Tey and Rodrigo Caballero  
August 28, 2023 at 2:00 a.m. EDT

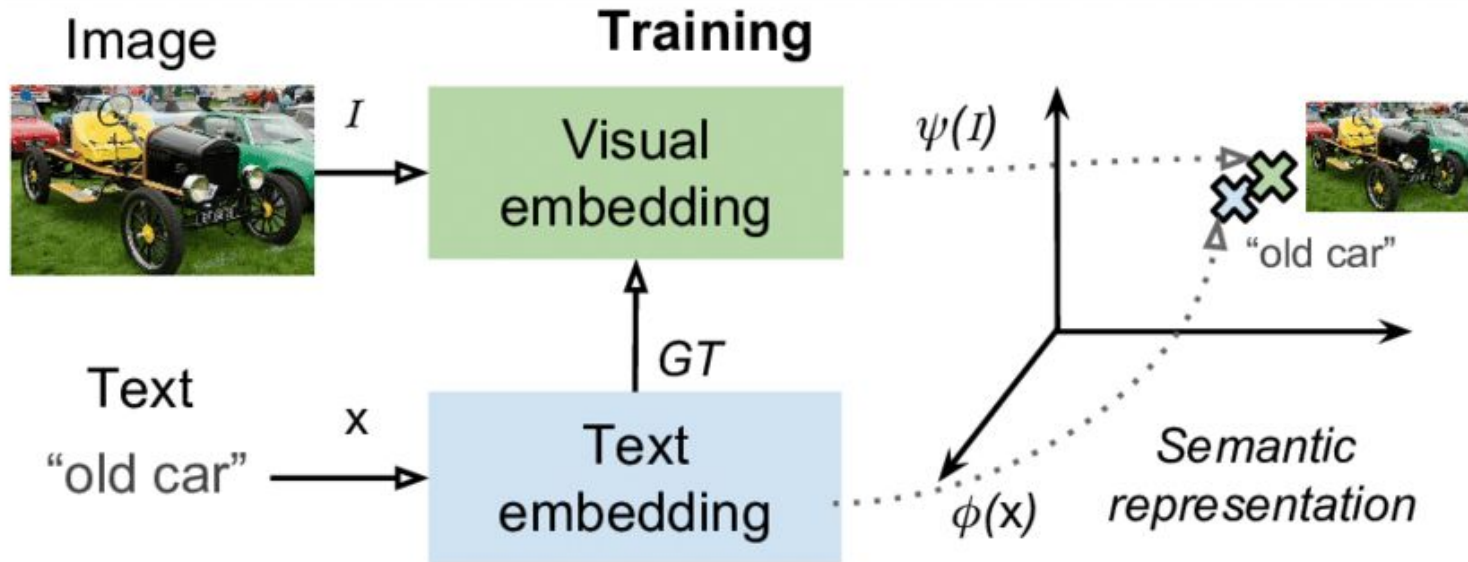


# Actualidad y futuro de NLP

# Introducción a Multimodalidad



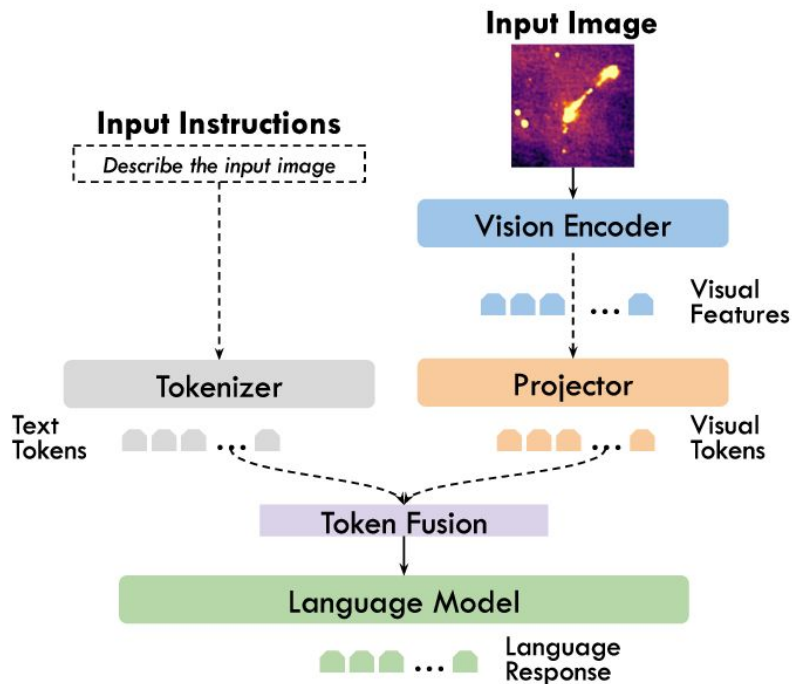
La multimodalidad es la capacidad de un sistema de inteligencia artificial para comprender y combinar diferentes tipos de información, como texto, imágenes, audio o video, al mismo tiempo.



# Introducción a Multimodalidad



LLaVA (Large Language and Vision Assistant) es un modelo multimodal que combina un encoder visual con un LLM para que pueda entender imágenes y responder preguntas sobre ellas.



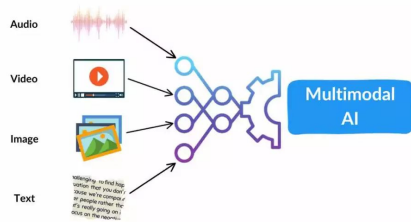


Link al Colab

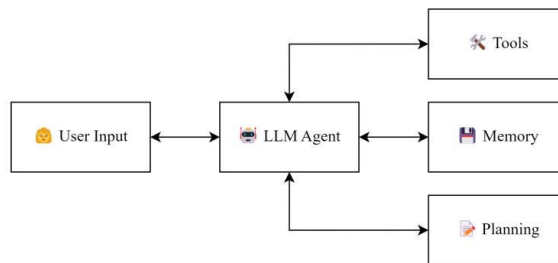


*LINK*

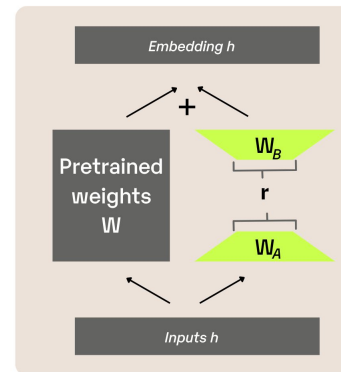
# Actualidad y futuro de NLP



Multimodalidad



LLMs + Agentic Systems



NLP Eficiente



# ¡Muchas gracias!